



Wye catchment soil phosphorus

A discussion paper prepared on behalf of Wye Salmon Association (WSA)

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Introduction

The recent report by Lancaster University (Withers et al., 2023), commissioned by the Environment Agency, discusses legacy phosphorus (section 4.1.3). An admittedly limited study of Wye catchment topsoils (to 300mm depth) fails to account for the calculated cumulative phosphorus input to the land. It is therefore reasonably speculated that the missing phosphorus has migrated to the subsoil or escaped into the wider environment. They suggest that such migration down the soil profile could be indicative of saturation, leading to enhanced P-loss to water (Chakraborty & Prasad, 2021). This is a key point which needs to be better understood by those concerned about the state of the river Wye. The first part of this note is therefore a brief review of the Chakraborty & Prasad paper.

A related aspect emerged while analysing historical EA water quality data in the Castlebrook sub catchment. It was striking how river nitrate levels dipped in the period 2002-3, just as we have previously observed in parts of the Garren sub catchment (Green, 2022a). However, in this case, the transient was equally apparent on the orthophosphate signal. This begs the questions: (a) is it real, (b) does it occur anywhere else, and (c) what does it tell us about the export of soil phosphorus? This is considered in the second part of this note.

Chakraborty & Prasad, 2021

This recent paper studies the long-term impact of repeated poultry manure (PM) application to agricultural land in Alabama. It specifically focusses on how different phosphorus forms are stratified within the soil profile. The land studied was a mix of pasture and arable, but with the latter primarily being under low tillage regimes to minimise soil erosion. Crops were predominantly corn, soybean and hay. Like pasture, low tillage systems (meaning little or no full depth ploughing) lead to significant stratification of nutrients within the soil. For example, it is well known (ADHB, 2022) that phosphorus from fertiliser application tends to remain in the top 150mm of pasture, but that on arable land, ploughing mechanically distributes nutrients over the top 250mm of soil. Thus, low-tillage land is closer to pasture than to conventionally ploughed arable land, resulting in increased stratification, with high phosphorus concentration close to the surface. Importantly, prior work is cited to support the assertion that high concentrations of P in surface layers are the main contributor to leaching and surface run-off.

Standard soil testing chemistries are different in the US to the UK. But the authors make the point that these standard tests take no account of P retention capacity and, just like our Olsen-P test, are not reliable indicators of P-loss risk. The authors prefer phosphorus saturation ration (PSR) as a P-loss risk indicator. PSR is defined as the molar ratio of extractable P, to the sum of extractable Al & Fe. It is therefore telling us about the fraction of available bonding sites (mostly on clay particles) which are occupied by phosphorus. A PSR value above 0.11 is considered to have a high P-loss risk.

The land studied had an average application history of poultry manure of 10 years and application rates ranged over the range 2 – 7 t/ha/year, equating to 26 – 92 kgP/ha/year. By comparison, poultry manure spread to the UK NVZ N limit would imply about 45 kgP/ha/year, or 66kgP/ha/year under COGAP, so this work is highly representative of the Wye catchment in that respect.

Core samples were taken at four different depths down to 450mm and analysed using a sequential fractionation procedure utilising DI water, NaHCO_3 , NaOH , HCl . Following other workers, the results were then classified as follows:

- Highly reactive P (HRP) representing loosely bound P extractable by water and NaHCO_3 (the latter similar to our Olsen-P test). This is the immediately plant-available fraction and by implication is also the most readily leachable fraction.

- Moderately reactive P (MRP) representing P extractable by NaOH and HCl. This generally requires breakdown by soil microorganisms to make it plant available.
- Non-reactive P (NRP) representing the balance of total phosphorus. This fraction is strongly bonded to soil particles, predominantly clay, and in the absence of bulk soil transport is not mobile.

Figure 1 below summarises the fractionation results from the paper¹:

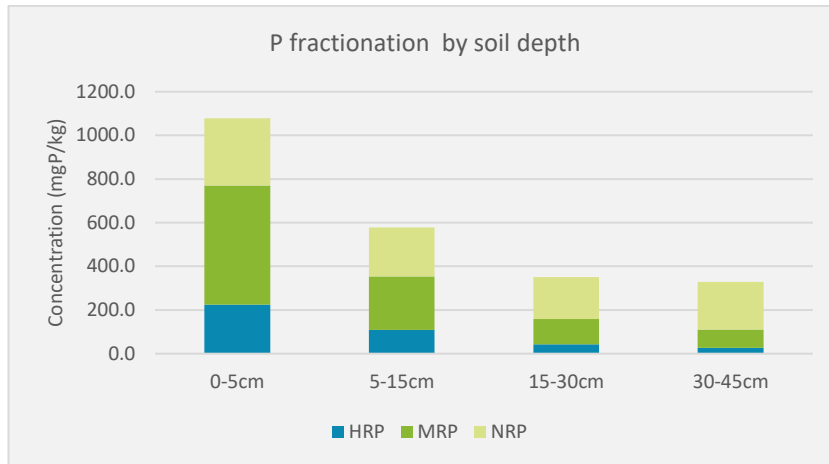


Figure 1: average distribution of phosphorus fractions at different soil depths, expressed as concentration (mgP per kg of soil). HRP, MRP & NRP are as defined above.

The two upper soil layers are dominated by HRP and MRP, which is why these layers dictate run-off and leaching to surface water. Note the levels of HRP. If we accept a rough equivalence to Olsen-P for this fraction, the top 5cm of soil is at Index 8 and the next 10cm below this is at Index 6². Mobile forms of phosphorus fall away rapidly at deeper levels, which become dominated by NRP. This implies that these soil depths are of much less concern for run-off and leaching. These results also suggest that the standard practice of sampling Olsen-P at 0-15cm on pasture, conceals the fact that the top 5cm of soil could have a much higher concentration of mobile fractions than the 0-15cm average would suggest.

Figure 2 below shows the PSR value measured for each depth range³:

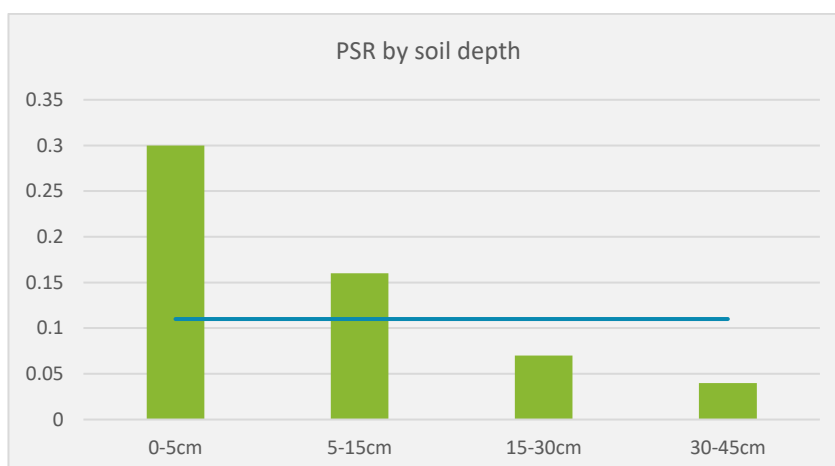


Figure 2: average phosphorus saturation ratio (PSR) at different soil depths. At PSR values over 0.11 (blue line), we expect to see enhanced leaching rates.

¹ This is the author's graph using the tabulated data from Table 3 in Chakraborty & Prasad, 2021.

² We found pasture in the Garren sub catchment ranging up to Index 7 (Green, 2022b).

³ This is also the author's graph, being an adaptation of Figure 4 in Chakraborty & Prasad, 2021.

In this graph we see that PSR at 0-5cm and 5-15cm substantially exceeds the environmental threshold for P-loss risk, whereas the lower layers do not. Stating this a different way, the soil down to 15cm has reached saturation; the NRP bonding sites on the clay particles here are more or less filled up. This forces additional applied phosphorus to remain in mobile forms, which has two consequences. It drives leaching and run-off, and it drives transport to the deeper layers which are not saturated and have the ability to accept more phosphorus as NRP (the point made by Lancaster Uni.).

Most manure within the Wye catchment is being spread on pasture, whether ruminant or poultry. Hence this paper is highly relevant. Our own soil survey in the Garren sub catchment and data presented by Lancaster Uni. (Withers et al., 2023) suggest that arable land, whilst being generally above agronomic optimum, has been relatively stable in recent years. Livestock farmers on the other hand are swimming in manure, be it slurry, sheep manure or poultry manure, and Olsen-P on pasture seems to be on a long-term increase.

We should not expect to translate numbers from this paper. The soils of Alabama are not exactly the same as the Wye and much of the latter has been receiving poultry manure for twenty, not ten, years – albeit at very similar application rates. The general principles, though, of saturation and stratification probably do transfer.

When Withers et al. calculate the 150-year accumulation of legacy-P for the Wye and can't account for it within the topsoil, it seems eminently reasonable to speculate that much of this has migrated down the soil profile because of surface layer saturation. This is entirely in line with the findings in Alabama.

However, what this paper also teaches, is that the top two inches of soil is responsible for most of the run-off and leaching. This is the layer we need to worry about. The deep-lying legacy-P is (a) predominantly non-reactive, (b) possibly well below saturation, and hence leaching much less than the surface layers.

The implication of this paper is that it is not necessary to draw down all legacy-P in order to have a meaningful impact on river RP values. We just need to reduce P concentration in the top few inches of pasture soil.

An obvious comment at this point is that this is all rather theoretical. But we did once in the past do a real-world trial on this in several areas of the Wye catchment, albeit not from choice. I refer to the 2001 foot and mouth outbreak.

Historical trends in water quality

This section admits to a degree of serendipity. In our report on the Garren sub catchment (Green, 2022a), we noted a significant dip in nitrate concentrations measured by EA on the Garren during 2002-3, which we speculated could be attributable to the 2001 foot and mouth outbreak. During more recent work on the Castlebrook sub catchment a similar pattern was found; moreover, it was present on the orthophosphate data as well. The significance of this with respect to the Chakraborty & Prasad paper, and by implication to the Withers et al. report, was striking. Could the rivers really respond this quickly to a reduction in manure load? First, a recap of the history.

Foot and mouth disease was first identified in England in February 2001. From there, case numbers rocketed, reaching a peak around the beginning of April and then slowly subsiding, with the last case notified on 30-September that year. In total, 2000 infected premises, plus a further 8000 associated premises were culled, resulting in the deaths of 6 million cattle and sheep. DEFRA archives show that Gloucestershire, Herefordshire, and the Welsh Borders suffered a large number of cases. Detailed on-line data from this period is sparse, but press reports give us some relevant information. A [Guardian](#)

[report](#) lists all infected premises as of 26-March – about one third the way through in terms of case numbers. Within this we already find a heavy concentration comprising five farms in the upper Garren sub catchment and two in the adjacent Gamber. In another [Guardian report](#), we find that unfenced sheep in the Forest of Dean were a major transmission vector, resulting in their culling in early April – which unfortunately did not halt the spread of the disease around the periphery of the forest. Thus, it seems highly plausible that the Castlebrook sub catchment was one such area (unlike today, the area was predominately pasture back in 2001).

Control measures were in place from the onset of the outbreak until January 2002. As well as a ban on movement of livestock, there was a ban on movement of manure. It was permissible to spread manure within an uninfected farm, but permissible spreading methods were limited. However, it's difficult to imagine many farmers seeing slurry spreading as a priority in 2001. And by 2002 many of the herds were gone. Difficult times.

The upshot, though, is that there would have been a massive reduction in manure load on pasture within infected areas, beginning in 2001. And this would have persisted until the reestablishment of culled herds a considerable period later.

Returning to the EA water quality data, figure 3 shows the nitrate and orthophosphate trends for the for Castlebrook.

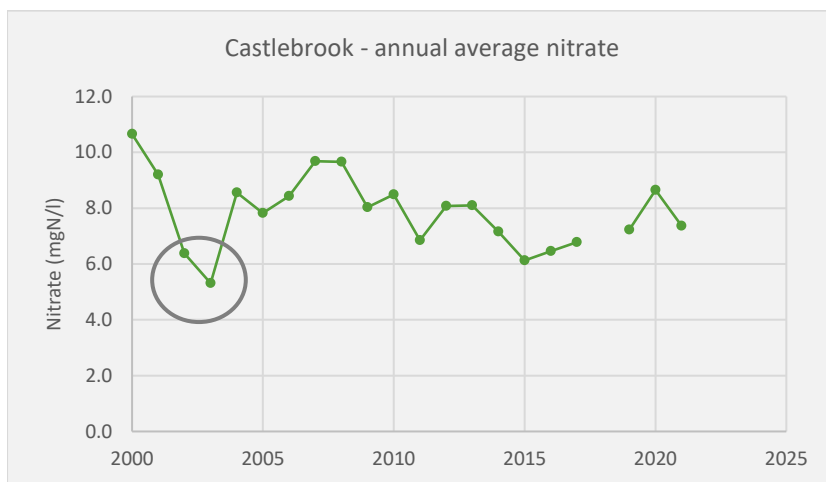


Figure 3a: annual average nitrate concentration on Castlebrook at Walford Church (EA data)

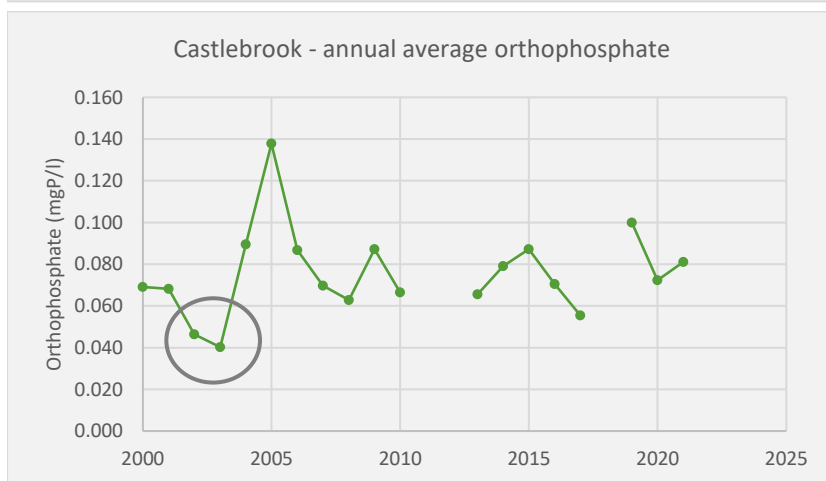


Figure 3b: annual average orthophosphate concentration on Castlebrook at Walford Church (EA data)

Note that on both charts we see unusually low values in 2002 and 2003 (circled). A similar pattern emerges on the Garren, as shown in figure 4.

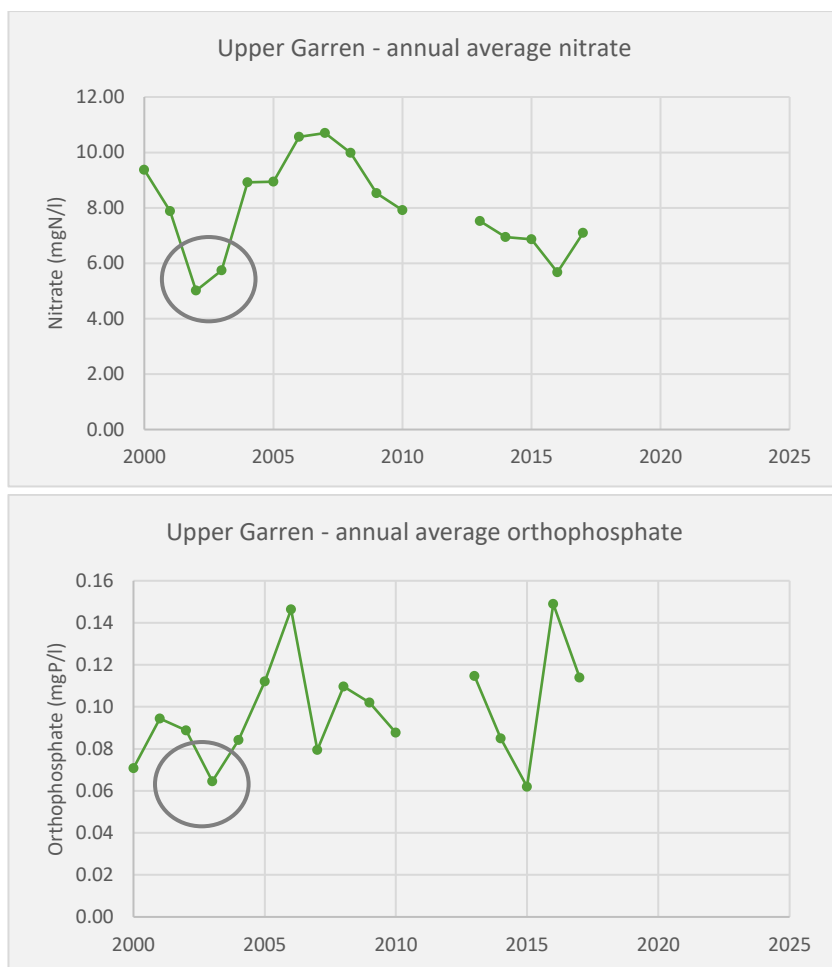


Figure 4a: annual average nitrate concentration on the upper Garren at Llangunnoch Bridge (EA data)

Figure 4b: annual average orthophosphate concentration on the upper Garren at Llangunnoch Bridge (EA data)

Here, the nitrate signal dips in both 2002 and 2003, but the orthophosphate signal is a little less convincing. However, it is not sensible to make subjective judgements on such charts. An objective statistical test would be far better. What we need to determine is whether the low results in 2002 and 2003 are merely statistical outliers, or whether they represent a genuine shift. We cannot use annual averages to do this because the number of samples per year is not always the same, so we shall analyse the underlying base data (this is mostly, but not always, monthly). We shall use a single-sided Student t-test to calculate the probability that data recorded in 2002-3 belong to the same population as data recorded in 2004-6.

Table 1 below shows the results at several, arbitrarily selected locations around the lower Wye catchment, including the two discussed above. It is clear that the pattern of anomalous low values for nitrate during 2002-3 is not restricted to Castlebrook and the upper Garren. It occurred on the Cage Brook, on the Gamber, on the Frome, on the Dore and on the Worm. Reductions in nitrate signal during 2002-3, compared to 2004-6, ranged from 18% to 49%, with an average across these sites of 31%. In the *Confidence* column, we show the probability that this is a genuine change⁴. These are extremely high values and show that the fall in nitrate was a real effect at these locations. We find corresponding dramatic reductions in orthophosphate, ranging from 11% to 59% and with a mean value of 34%. Confidence values are similarly high, again indicating incontrovertibly that these were real changes in 2002-3.

⁴ The t-test calculates the probability of the null hypothesis – that these reductions have occurred by chance. Here we present 100% minus this value, to present the probability of this being a genuine change.

Location	Nitrate		Orthophosphate	
	Reduction	Confidence	Reduction	Confidence
Cage Brook at Cage Brook Br.	18%	91%	36%	99.8%
Castlebrook at Walford Church	29%	95%	59%	97%
Gamber Brook at Kilbreece Br.	25%	89%	25%	91%
Garren Brook at conf. with R. Wye	33%	98%	19%	68%
Garren Brook at Llangunnoch Br.	43%	99.6%	33%	91%
R. Frome at Longworth Br.	46%	99.5%	11%	93%
R. Frome at Rowden Abbey Br.	49%	99.9%	33%	86%
R. Dore at conf. with R. Monnow	27%	85%	28%	97%
R. Dore Elm Br., Ewyas Harold	18%	93%	34%	95%
Worm Brook at conf. with R. Dore	28%	94%	47%	99.3%
Worm Brook at Wormbridge	21%	88%	45%	96%
Average	31%		34%	

Table 1: fractional reduction in average nitrate and orthophosphate concentrations during 2002-3, compared to 2004-6 (EA data). Confidence is calculated as 100% minus the P value from a single-sided Student t-test.

Obviously, a strong correlation does not prove a causal relationship. However, it is very difficult to see what else caused this anomaly, which hasn't been repeated in the intervening years. 2002-3 doesn't throw up any extreme rainfall, or river flow rate pattern which might otherwise explain the data. The most likely cause remains the disruption to the agricultural sector caused by the 2001 foot and mouth outbreak.

Assuming this to be the case, then the clear implication is that if manure deposition on pasture is significantly curtailed, then the rivers respond immediately and profoundly to this change. This reinforces the message from Chakraborty and Prasad, that export to surface water from pasture is dominated by the top few inches of soil. It is not necessary to run down the entire legacy P burden in order to reduce phosphate in the rivers.

Discussion

Much has been made of the legacy-P issue relating to the eutrophication of the Wye catchment. The clear inference is that legacy-P stored in the soil needs to be drawn down in order to reduce exports to the rivers. That such reduction in legacy-P will take decades to achieve at typical crop off-take rates is inescapable, and the recent input from Withers et al. that legacy-P has in fact been building up since the mid-19th century (their figure 4b) appears to add to the woes. But beware that a simplistic "single reservoir" model of this appears to be misleading and overly pessimistic.

Chakraborty and Prasad show how different phosphorus forms are stratified within the soil profile of permanent grassland and low tillage arable land. Mobile forms of phosphorus are heavily concentrated in the top few inches of soil and, hence, it is this zone which dominates the quantity of P exported to surface water, either by leaching or by run-off. Repeated application of organic fertiliser (in their case poultry manure, but the same logic would apply to slurry and digestate) ultimately causes P-saturation in these surface layers. This forces the applied P to remain in mobile forms which (a) permeate down the soil profile and (b) increase export rates.

The soil PSR parameter is proposed as a useful measure of saturation and thus of environmental export risk. It is shown how PSR reduces with increasing depth, and a reasonable inference would be that the deeper, unsaturated layers remain nett absorbers of P. The falling concentration of HRP and MRP with increasing depth can be seen as supporting this view. As water permeates down the soil profile (ultimately en-route to the river), it progressively loses P to unsaturated zones, depositing as NRP. Thus, the HRP and MRP in the deeper, unsaturated layers do not represent emissions from these layers, but are surface-originated P in-transit. This latter point is particularly important.

This work is suggesting that the deeper, unsaturated soil strata which contain much of the legacy-P are not in fact significant exporters and indeed are likely to be nett absorbers. Hence, the deep-lying legacy-

P is unlikely to be driving export to surface water. Rather, the problem to solve is the saturation of the upper soil layers – which is a much more tractable challenge. A simple calculation shows that the average P load in the top 5cm of soils studies by Chakraborty & Prasad is about 500 kgP/ha. Which represents many decades of typical pasture draw-down rates. But we need to be clear about the objectives. The Wye catchment is agricultural land, which will always require moderately elevated P levels to maintain yields. Zero is not the target, any more than zero RP is required in the rivers. The surface soils need to be brought out of saturation and down to agronomic optimum level. Moreover, since the upper layer, when saturated, has most of its P in the form of HRP & MRP, we are not reliant solely on crop off-take. Natural drainage of mobile P forms into the lower soil is also working for us. Thus, Chakraborty & Prasad is suggesting that surface water export might respond reasonably rapidly to an appropriate reduction in application rates for P containing fertilisers. It is certainly not necessary to draw down all legacy-P.

The water quality data from immediately after the 2001 foot and mouth outbreak tells a compelling story of how the streams of Herefordshire responded to a profound reduction in manure spreading. Although we cannot prove a causal link, it is difficult to see an alternative explanation for this very robust correlation. Taken at face value then, averaged over two years, we find that RP levels in the water bodies examined above dropped by one third – and did so more or less immediately. This is surely a very encouraging result which strongly supports the implications of the Chakraborty & Prasad paper.

A reasonable inference is that, if a similar reduction in manure loading could be achieved today, then similar RP reductions would be observed in the rivers. Potentially, the Wye could be returned to acceptable phosphorus levels in five years – not fifty.

Conclusion

- Chakraborty & Prasad teaches that:
 - Repeated manure application can cause P-saturation in upper layers of soil in pasture and low tillage scenarios.
 - These surface layers are in general responsible for the bulk of export to surface water, but saturation increases export rates as well as driving the migration of P to lower soil layers.
 - Most P in lower soil layers is not mobile and is not driving export to surface water.
 - PSR appears to be a very useful parameter for assessing P-loss risk.
- By implication:
 - It is not necessary to draw down all legacy-P to reduce surface water export.
 - The primary objective is to eliminate saturation in the top few inches of soil (on pasture).
 - River RP levels could therefore be significantly reduced in years, not decades, if adequate reductions in manure application rates could be achieved.
- EA water quality data from immediately after the 2001 foot and mouth outbreak strongly support the view that immediate and significant reductions in nitrate and orthophosphate concentrations can occur when manure spreading is seriously curtailed.

References

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Abbreviations

HRP	Highly Reactive Phosphorus
MRP	Moderately Reactive Phosphorus
NRP	Non-Reactive Phosphorus
P	Phosphorus
PM	Poultry Manure
PSR	Phosphorus Saturation Ratio
RP	Reactive Phosphorus